



The Art of Sampling Transcript

So, as I was discussing about inferential statistics, the very first lesson that we have right now or what we are going to study right now is about sampling. How do we sample? What exactly is the mechanism of sampling? There are various different analogies that anyone can bring in, but one of the most easiest ones that I could relate to is, let's say if you are making a pot full of soup and you would want to know whether the soup is good or not, you can do two things. One, you can actually take the entire pot of soup and consume it and then decide whether it is good or not.

Second thing is, you take a tablespoon of soup and then taste it and decide whether the soup is really good or not, right? So, the process of taking a spoon of soup is the sample. Soup or a tablespoon of soup is exactly a small sample of the entire pot. Pot is the entire population.

So, this is one of the closest analogies that I can draw to help you understand what exactly is a sample and what exactly is a population. Now, I also talked to you about two different things. One parameter and the estimate, right? In this analogy, if you think of, the parameter that we are trying to learn about is how good is the soup, right? And for that, we have the entire soup and we take an estimate using a tablespoon of soup, right? Now, to do the entire analysis on the entire population, why is it not very feasible? There are two different reasons and two most important reasons.

As I told you, the very first thing is, it is very time consuming. It might also incur a lot of resources in terms of money, time, xyz, whatever it is, right? And also, it might not be very feasible. Think of, if you have to investigate entire 140 billion people living in the country, it might not be very, very, I mean, not the way you should be thinking about, right? The convenient way is to take a sample and then decide about the entire population using that sample.

So, in this entire episode of things, we are going to learn about sampling. What exactly is a population and sample? I told or I gave you the example of a pot full of soup and one simple tablespoon of soup, right? Similarly, let's say, if you would want to study about different types of university graduates in India, what are their average GPA, CGPAs and so on, you could take the entire students who are studying in various different universities in India, that's a population. And if you take 1000 students from a specific university in India, that's a sample.

So, a sub part of a population is a sample. Now, the next thing is, there are two different types of sampling that we have. The entire mechanism has two different types.

One, which is called as probability sampling. The other one is called as non probability sampling. Probability sampling is the one wherein every single individual of the



population or every single thing of the population has some chance of being represented in the population.

So there is some non zero chance of being representative of the sample. Why non probability sampling is something which is very convenient to devise this type of sampling based on your own convenience, let's say, it might not be very convenient for you to go ahead and take samples from Rajasthan, Gujarat and Kerala. So what you might do is take samples which are closer to us or people who are closer to us, right? So probability sampling is more about bringing more randomness, while non probability sampling is about convenience, it's whatever is more convenient to us, whatever way is more convenient to us, we would prefer that way.

But non probability sampling, on the other hand, it brings in a lot of biasness as well. Now, why do I say biasness, let's say, for example, if you're trying to estimate average height of people living in India, in that case, if you take a sample only from a place wherein people are short heighted, then in that case, your entire analysis would be biased because let's say, if the people are short heighted, your average of all those peoples in the sample might come out to be five, while the entire India's average height could be 5.5, which is a huge difference. So your output based on a biased sample would always be biased, would also be biased, that's what it means.

Now, different types of probability sampling, there are various different ways of taking samples using probability sampling. One of the ways is simple random sampling. Basically, it's about bringing randomness by randomly selecting the samples or randomly selecting people of the entire population or randomly selecting objects within the entire population.

So that's what is a random sampling, wherein every individual has equal chance of being chosen. Let's say, for example, drawing names from a hat, right? So most of us would have played this game wherein you have chits or some sort of names written in chits, and then you have to pick some chit, right? So every single individual, if we imagine or assume that as a part of names written in those chits, then every chit has same or equal chances of being picked. So that's simple random sampling.

Stratified sampling. Now, before we do random sampling, we can also introduce a concept of strata, strata as in groups. So let's say if we have to estimate average height of people living in India, we can divide the entire population into groups, which could be states or which could be regions.

Now you can take samples from each region or you can take same amount of samples from each region or each state. So we first divide the entire population into groups and those groups we take samples from. So that's what is called as stratified sampling.



Then we have systematic sampling. Systematic sampling as in, let's say we decide to take samples every five minutes or we divide to take a sample from every fifth individual that we meet. So if there is a strategy involved, if there is more randomness involved based on certain time or based on certain number after a particular frequency, all of those things are called as systematic sampling.

Then we have cluster sampling. What does cluster sampling mean? Cluster sampling is when you divide the entire population into different stratas or clusters or groups, but you also randomly select the cluster from where you have to take the sample. So there are two pieces of randomness that you are bringing in here.

First, the first randomness is to select the cluster itself, randomly select the cluster. So let's say if I have to estimate average height of people living in India and I decide to divide the entire population of India into North, East, West, South and these are the four different regions that we have. Now the very first thing that I can do is to select randomly out of these four regions which region would I be taking sample from.

So that's the first random step. Second is once I have decided which region it is, then I'll go ahead and implement simple random sampling. So this is a two step process.

That's why it is called as cluster sampling because you have clusters involved. Now next is about very practical sampling which we called as non-probability sampling and this I told you majority of the things are convenience driven. So the very first sampling is convenience sampling, selecting whosoever is easily reachable or accessible.

That's the convenience. Second is quota sampling. Quota sampling as in, let's say you divide the entire population into groups, but then you also say that I will conveniently select first 50 in those groups.

So let's say we divide the entire population into two genders, male and female. And then we say that we will conveniently take first 50 male candidates and interview them, first 50 female candidates and interview them. That is quota sampling.

Then we have snowball sampling. What does it mean? Let's say you take sample from one individual and then you draw a chain. Chain as in, you ask the individual to bring in more people, maybe 10 more people, and then you interview those 10 people or you take sample from those 10 people and then you ask them to bring in 10 more.

So there is a chain of things which is involved here. That's what is called as snowball sampling. Now, how big or small should be a sample? And how do we decide how big or small the sample should be? This entire thing is called as Goldilocks problem and not too big, not too small.



So that's a trade off, right? And all of these things are decided by three different metrics that we have. One confidence level or we generally call it as level of significance. The second is called as margin of error.

The third is called as population variability. So the very first thing, confidence level. How sure do you want to be? If you would want to be 95% confidence, then obviously you would require larger sample.

And this is very intuitive if you could think of. If you want to be more confident about your population, then you would have to take larger sample size. The second thing is margin of error.

How much inaccuracy do you want to tolerate in the entire sampling process? So if you want least amount of inaccuracy, then you'll have to take a larger sample. Then if your population is very variable, I mean, there are more variability in it, let's say the average height of, sorry, not average, but height of people living in India could vary from 2 to 8. Now if there is this much amount of variability, then obviously you will have to take more sample to estimate the correct average height. Now if the statistic itself that you are trying to take from the population has a very little variability, then in that case, your work can be done with even a very small sample.

So that's what variation means. If you have more variation, take larger sample. If you want least amount of errors, take larger sample.

If you want to be more confident, then take larger sample. So that's why it is called as how big should be the sample. Now this is about an example of bias, which got introduced in one of the presidential elections in US in 1936.

It seems that there was a statistical survey which was designed to take survey from or take sample from somewhere around 2.4 million people. So they took a sample of 2.4 million people, asked them about who do you want to see as the presidential candidate. And what they found out was, after everything happened and they said that using the entire strategy, sampling these 2.4 million people, asking them about who would win, they estimated that individual X is going to win.

But what happened was individual Y won. So this entire bias came in because they took the entire sample from people who are owning cars and who had telephone. All of those people who were owning cars or had telephone, they represented a small subsection of the entire society or the entire population.

So this is the bias you can think of. You can imagine what got introduced there. This is the bias because all those people wanted X to win, while the entire larger section of people, which were not included in the sample, they wanted Y to win.



So this entire thing also tells us about how good the sample should be to estimate the right population statistic. So your sample should very well be good representative of the entire population. So in this section, we have learnt about a lot of things starting from sample and then we went into different types of sampling.

We learnt about inferential statistics as well as a whole. And then we learnt about the bias, which might get introduced because of different types of sampling that we take. So with this, I would next move on to one of the estimation techniques, which is called as point estimation.

So let's meet in that video. Thank you.

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